



Education to Eternity

JAIDEV EDUCATION SOCIETY'S J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of CSE

"A place to learn; A chance to grow"

2025-2026(Odd Semester)

MSE-2 EXAMINATION



॥ ज्ञानं सत्यं धर्मं ॥

Course: B.Tech. In Computer Science and Engineering Year/Semester: 6th Semester (3rd Year)

Subject: STQA Subject Code: CS6T002

Max Marks: 20

Date: 30/03/2026

Duration:- 1 Hr.

Instructions to the Students:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:

CO1: Student will be able to Understanding the Basics of Software Testing and Its Importance.

CO2: Student will be able to gain a thorough understanding of unit testing and Debugging Techniques.

CO3: Student will be able to Applying System Integration Testing and Test Design Techniques.

CO4: Students will learn to categorize and perform various system tests.

CO5: Student will be able to Performing Acceptance Testing and Ensuring Software Quality.

SECTION A

1. All Questions are compulsory.
2. Answer in one word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Control Flow Testing belongs to which type of testing?	L1/CO3	1
Q.2	Why are nodes used in a Control Flow Graph?	L2/CO3	1
Q.3	What is meant by a variable definition in data flow testing?	L1/CO4	1
Q.4	Why is functional testing important in software systems?	L2/CO5	1

SECTION B

1. All Questions are compulsory.
2. Answer in one two three sentence.

Q.5	How does Control Flow Testing help in identifying logical errors?	L2/CO3	2
Q.6	Identify the path in CFG for the condition if ($x > 10$) when $x = 15$	L3/CO3	2
Q.7	Why are data flow testing criteria required?	L2/CO4	2
Q.8	A banking application must handle 5,000 users accessing the system simultaneously. Which testing type should be applied?	L3/CO5	2

SECTION C

Q.9	Analyse the importance of identifying paths in a Control Flow Graph during software testing.	L4/CO1	4
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30



॥ वाचं धेनुमुपासीत ॥

Consider the following program segment:

Q.10

```
int x = 0;
2. if (a > b)
3. x = a - b;
4. else
5. x = b - a;
6. print(x);
```

L4/CO3

4

Analyse the definition and uses of variable x and identify the definition-use (DU) pairs.

Q.11 Analyse the differences between Load Testing and Stress Testing.

L4/CO4

4